ECON100B - SECTION 3

The Long Run Economy

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 1 has been released! Due Friday, 2/13

Agenda

- Growth Rates
- The Long Economy
- PSET problems

Question of the Day

1. PSET 1, Q1b: What exactly does investment spending create? Explain?

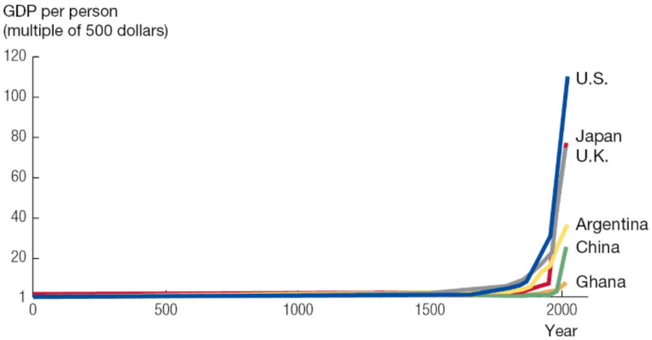
2. What’s your favorite building on the UC Berkeley campus?

The Long Run Economy

The Very Long Run

FIGURE 3.1

Economic Growth over the Very Long Run in Six Countries



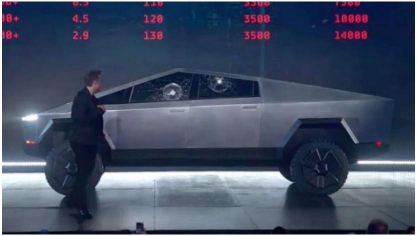
Source: The Maddison-Project, ggdc.net/maddison/.

How did this happen?

1700

1910

2020



Growth Rates = Percentage Changes

Recall our percentage change formula from last class:

$$(x_{t+1} - x_t)/x_t$$

The growth rate of an economy can be calculated using this formula!

$$(y_{t+1} - y_t)/y_t = \overline{(g)}$$

For example, growth from 2024 to 2025:

$$(y_{2025} - y_{2024})/y_{2024}$$

Growth Rates = Percentage Changes

We can rearrange this equation to project our future income/output Y, if we know the growth rate and current income/output:

$$y_{t+1} = y_t\left(1 + \overline{(g)}\right)$$

For multiple time periods, with constant growth rate g, we get:

$$y_{t+1} = y_0\left(1 + \overline{(g)}\right)^t$$

Working backwards to growth rates:

Say we know the GDP of a country in year t and the GDP of a country in year 0. how do we find the growth rate?

$$y_{t+1} = y_t\left(1 + \overline{(g)}\right)^t$$

$$y_{t+1}/y_t = \left(1 + \overline{(g)}\right)^t$$

$$\bar{g} = \left(\frac{y_t}{y_0}\right)^{1/t} - 1.$$

Some additional properties of growth rates

Suppose two variables x and y have average annual growth rates of gx and gy: then the following rules apply:

1. $z = x / y$, then $g_z = g_x - g_y$.
2. If $z = x \times y$, then $g_z = g_x + g_y$.
3. If $z = x^a$, then $g_z = a \times g_x$.

Why do these rules work?

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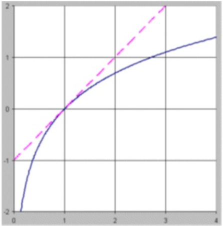
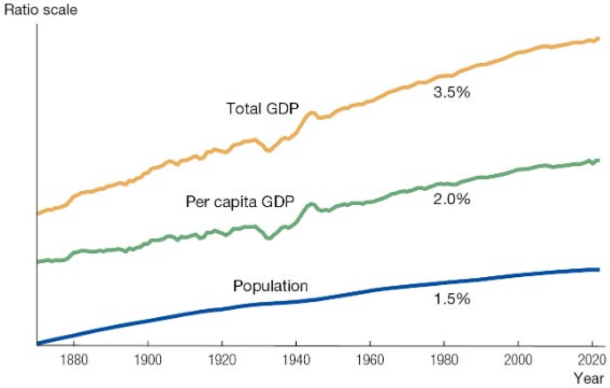


The second property of growth rates

If total GDP in an economy equals the product of per capita GDP and the population, the growth rate of GDP is the sum of the growth rates of per capita GDP and population



The Very Long Run

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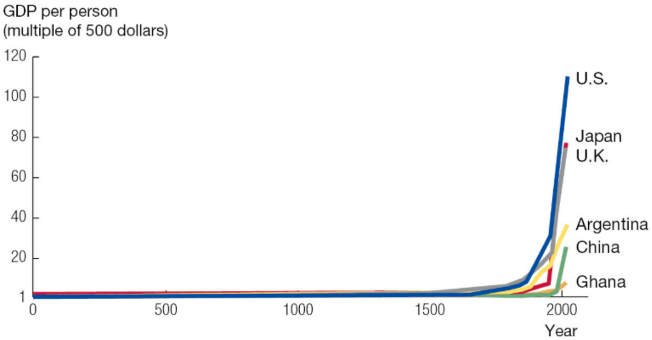


Source: The Maddison-Project, ggdc.net/maddison/.

The Rule of 70

How many years does it take for income to double in a country? Estimate with the rule of 70, where it takes 70/g years for the income to double!

The Rule of 70

How many years does it take for income to double in a country? Estimate with the rule of 70, where it takes 70/g years for the income to double! Proof:

$$y_t = 2y_0 = y_0\left(1 + \overline{(g)}\right)^t$$
$$2 = \left(1 + \overline{(g)}\right)^t$$

I see an exponent … so let's a log!

$$ln(2) = t * ln\left(1 + \overline{(g)}\right)$$
$$t = \frac{ln(2)}{ln(1 + \mathrm{g})}$$

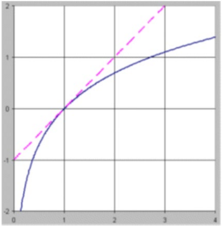
The Rule of 70

How many years does it take for income to double in a country?

$$t \approx \ln(2)/g$$

because $\ln(1+g) \approx g$

Recall from last class:



The Rule of 70

What the rule of 70 states:

The number of years it takes an outcome, y to double is roughly equal to 70 / g

If y grows at 2% per year, then it doubles about every 70/2 = 35 years

Why?

Recall $\frac{\ln(2)}{g}$ where $\ln(2) \sim 0.693$

The Rule of 70, Concluded

What the rule of 70 tells us:

- It doesn't matter the size of y: we can tell how long of a time t it will take something to double only by knowing growth rate g
- Small differences in growth rates can lead to pretty different outcomes over time!

- Consider a growth rate of 2 percent vs. four percent: With a growth rate of 2 percent, it takes (70/2) ~ 35 years for an economy to double in size. With a growth rate of 4 percent, it takes ~17.5 years for an economy to double in size. Therefore, an economy that has a growth rate of 4% compared to 2% will be 4x bigger in 70 years!

The rule of 70 concluded

Consider a growth rate of 2 percent vs. four percent: With a growth rate of 2 percent, it takes (70/2) ~ 35 years for an economy to double in size. With a growth rate of 4 percent, it takes ~17.5 years for an economy to double in size. Therefore, an economy that has a growth rate of 4% compared to 2% will be 4x bigger in 70 years!

Suppose both economies start at a GDP of $10B

In 17.5 years: Economy A: $14.14B Economy B: $20B

In 35 years: Economy A: $20B Economy B: $40B

In 52.5 years: Economy A: $28.28B Economy B: $80B

In 70 years: Economy A: $40B Economy B: $160B

Essentially, we're comparing $2^2 \ with \ 2^4$

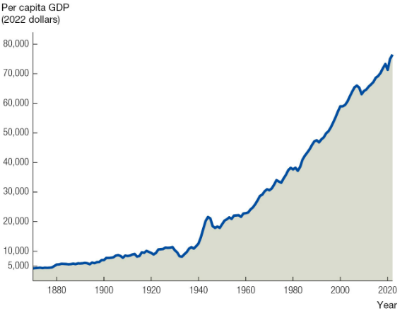
Ratio Scales

When plotted on a ratio scale: a variable growing at a constant rate appears as a straight line:

Why? We just observed that it takes the same amount of time for something to grow at a certain factor (say double or triple) regardless of where we’re starting from!

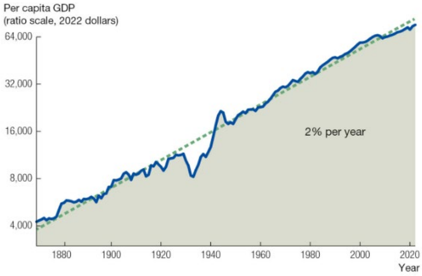
Ratio Scales

Per Capita GDP in the United States



Normal Scale

Per Capita GDP in the United States, Ratio Scale



Ratio Scale

PRACTICE

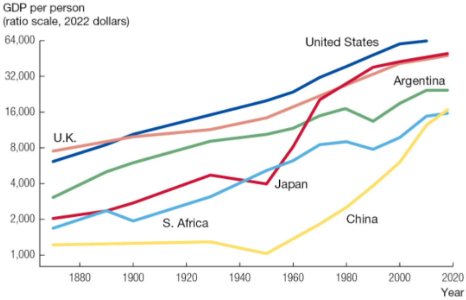
1. Suppose population growth is 3% every year. Using the Rule of 70, in roughly how many years does the population double?

2. If GDP per person (Y/L) grows at 3% per year while GDP (Y) grows at 3.75% per year, what is the population growth rate?

3. Given the function $y = k^{1/3}$ and a growth rate of 5% for k, what is the growth rate of y?

GDP growth rates across countries

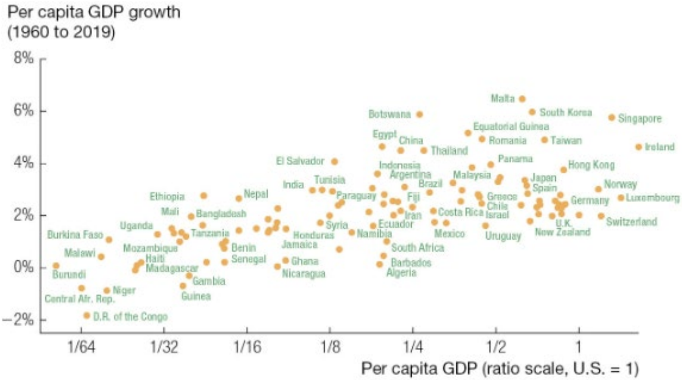
Per Capita GDP since 1870



Source: The Maddison-Project, ggdc.net/maddison/. Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

Growth Rates and GDP Per Capita

Levels and Growth Rates of Per Capita GDP



The costs of economic growth

Benefits

- Higher incomes and purchasing power
- Increased lifespans, reduced infant mortality
- Expansion of goods, services, and technology

Costs

- Shifts in jobs and industries that are popular within an economy
- Environmental degradation and resource depletion
- Growing income inequality across and within countries

PSET Practice Time

(c) [2 points] In the first half of ECON 100B, what is capital? Describe it in a few sentences. Maybe describe the economic meanings of the word that do NOT directly apply to our study of economic growth.

(d) [2 points] Recall that real GDP Y^R equals nominal GDP Y^N divided by the price level:

$$Y^R = \frac{Y^N}{P} \tag{1.2}$$

Take the growth rate of both sides of equation (1.2), and recall that the growth rate of the price level is inflation and denoted by a lowercase Greek letter "pi," π. If the growth rate in nominal GDP is 5% and inflation is 2%, what is the growth rate of real GDP?